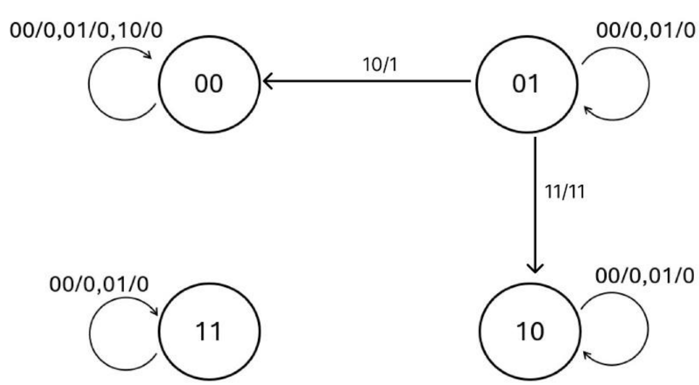
Lab 3

Q2: $D_A = \overline{X}A + XY$, $D_B = \overline{X}B + XA$, $Z = XB$:-

A. State table: -

Present State				Next State		
A	B	X	Y	D_A	D_B	Z
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	1	0	0
0	1	0	0	0	1	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	1	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	1	0
1	1	0	0	1	1	0
1	1	0	1	1	1	0
1	1	1	0	0	1	1
1	1	1	1	0	1	1

B. State diagram: -



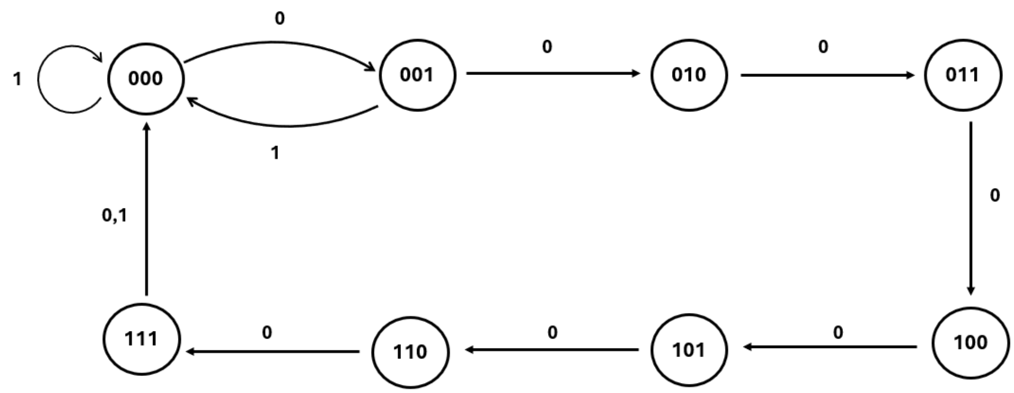
Q3: Derived Boolean expression of 3-bit counter using D Flip-Flop: -

$$D_3 = \bar{x}(A_1(A_2 \oplus A_0) + A_2\overline{A_1})$$

$$D_1 = \bar{x}(A_1 \oplus A_0)$$

$$D_0 = \overline{\bar{x}(A_0)}$$

State diagram:-



Q4: Starting from state 00 in the following state diagram, show the state transition sequence and output sequence for input sequence 0101101111: -

A/1. State Transition Sequence for input 0101101111: -

00 → 00 → 01 → 10 → 11 → 10 → 11 → 11 → 11 → 11 → 11

A/2. Output Sequence: -

0, 0, 1, 0, 1, 0, 0, 0, 0, 0

B. State table: -

Present State			Next State		
A	B	X	D_A	D_B	Y
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	1	0	0
1	0	1	1	1	1
1	1	0	1	0	0

1	0	1	1	1	0
1	1	1	1	1	0
1	1	1	1	1	0
1	1	1	1	1	0
1	1	1	1	1	0

Q5: State table of A sequential circuit has two flip-flops A and B, one input X, and one output Y : -

Present State			Next State		
A	B	X	D_A	D_B	Y
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	0
1	1	1	1	1	1

Q6: Draw the state diagram of the rotator, and the equations for each D flip-flop: -

A. Equations: -

Current states are Q_3, Q_2, Q_1, Q_0.

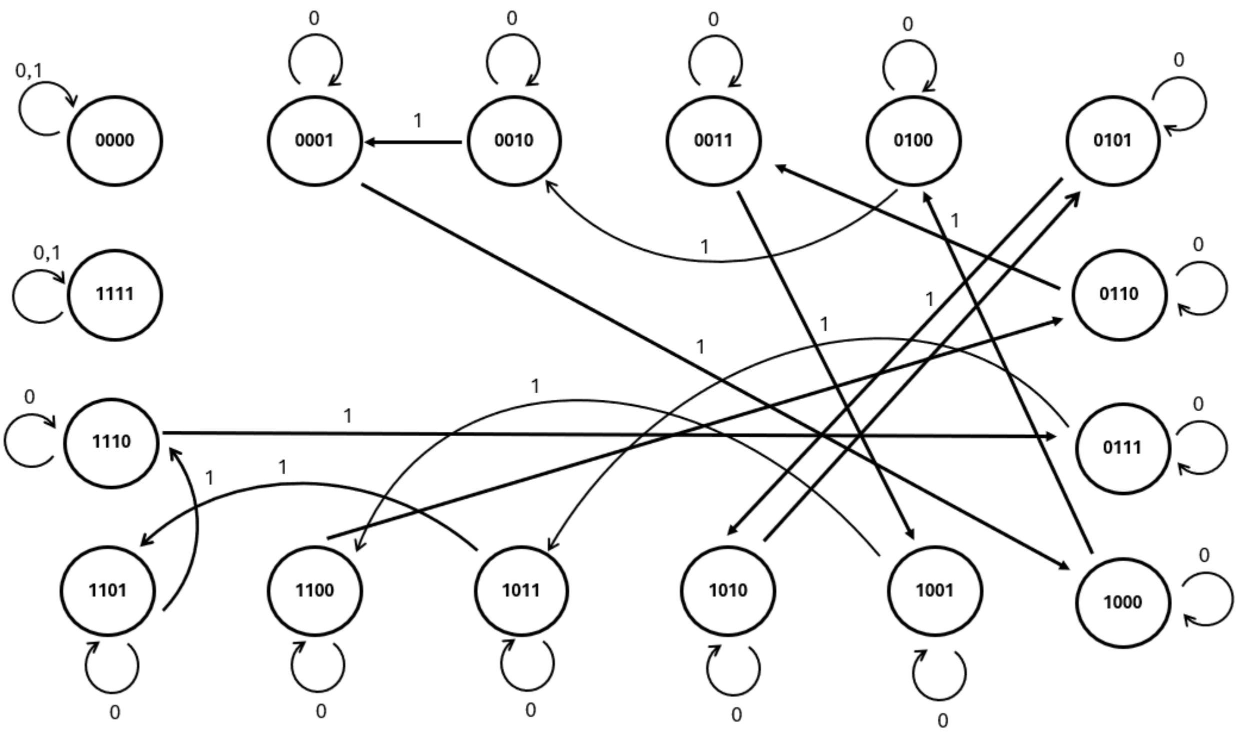
$D_0 = \bar{Y}X_0 + YQ_1$

$D_1 = \bar{Y}X_1 + YQ_2$

$D_2 = \bar{Y}X_2 + YQ_3$

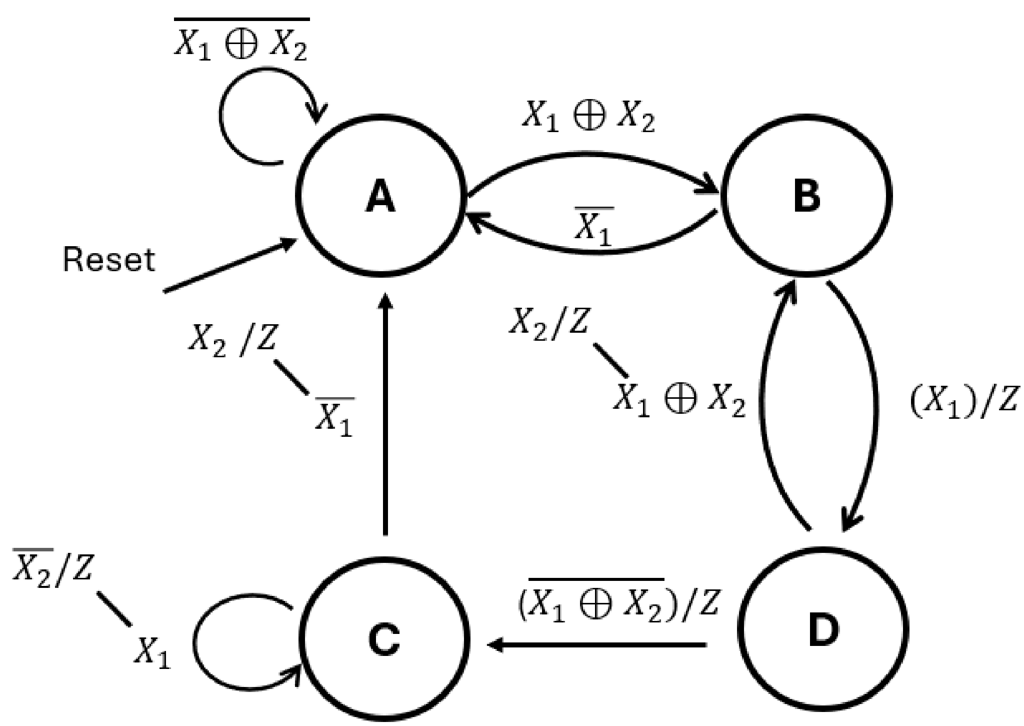
$D_3 = \bar{Y}X_3 + YQ_0$

B. State diagram: -



Q7: Work on the following state diagram: -

A. Draw the state-machine diagram: -



B. Perform state assignment, write down the Flip-Flop Input Equations and Output Equations, then optimise it: -

Present State				Next State		
A	B	X_1	X_2	D_A	D_B	Z
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	0	1	1
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	0	1	0	0	0
0	1	1	0	1	0	1
0	1	1	1	1	0	1
1	0	0	0	1	1	1
1	0	0	1	0	1	1
1	0	1	0	0	1	0
1	0	1	1	1	1	1
1	1	0	0	0	0	1
1	1	0	1	0	0	0
1	1	1	0	1	1	1
1	1	1	1	1	1	0